

ITU-T's internodal broadband signalling protocol

I G Jones

Broadband integrated services user part (B-ISUP) is an internodal signalling protocol defined by the International Telecommunications Union (ITU-T). Its purpose is to establish, maintain and release asynchronous transfer mode (ATM) connections in a public network, via the use of signalling messages. The initial release of B-ISUP was in February 1995, known as capability set 1 (CS-1). Since then, continued progress has been made extending the capabilities of B-ISUP with the development of CS-2. This paper provides an overview of the B-ISUP signalling protocol capabilities and describes its impact on the traditional Signalling System No 7 (SS7) architecture.

1. Introduction

B-ISUP [1] is an internodal signalling protocol, and as such can operate between nodes within either a national or international network. Like narrowband ISUP (N-ISUP) [2] it is based upon a common channel signalling concept. In common channel signalling, a dedicated signalling channel is used to transfer signalling messages between nodes. These signalling messages related to the control plane are used to establish, maintain and release user plane calls/connections. The exchange of signalling messages creates an association between the control plane and user plane. Once this association has been created, a call/connection is said to have been established and user plane data can be sent and received. When the call/connection is released the association between the control and user plane is then removed.

In an ATM environment, each virtual path (VP) can contain 65 535 virtual channels (VCs). These VCs are divided into two categories — VC identifier (VCI) values 0—31 are reserved for control and management plane functionality, while VCI values 32—65 535 are used for transferring user plane information. A single dedicated signalling VC (SVC), using VCI = 5, serves to transfer signalling information between the nodes involved in the call/connection. The signalling VC is preconfigured by management, i.e. permanent VC (PVC), and is used to establish, maintain and release the user plane connections, in real time. This concept is known as on-demand or switched connections. Depending on the configuration chosen, an SVC can either control only the user plane connections within its own VP, or alternatively may in addition control user plane connections within other VPs. The first type is known as associated signalling, where the SVC is always associated with its VP, while the second type

is known as non-associated signalling. Therefore a single signalling VC can control many thousands of user plane connections.

ATM connections are very flexible, allowing a wide range of their connection characteristics to be specified. Using signalling, end users can dynamically choose the bandwidth of their connections and, once established, have the freedom to then either increase or decrease this bandwidth. Also, a user can select from a range of transfer capabilities, including constant, variable, block transfer or available bit rate services. A user can also put constraints on the quality of service (QoS) offered (i.e. how many cells can be lost, specify the transfer and variation delay of cells), and select a connection topology, be it point-to-point or point-to-multipoint, etc. Therefore, to control these ATM connections, the signalling protocol has to be able to transfer this information in a flexible way, and the end-user terminals, plus the switches, have to process and understand this information.

2. Overview of Signalling System No 7 (SS7) architecture

Figure 1 shows the broadband SS7 architecture. As can be seen, the B-ISUP signalling procedures represent only one part of this overall architecture. Just like any other large complex system the SS7 functionality is divided into a number of sub-systems, each designed to perform a specific task. The B-ISUP signalling procedures establish, maintain and release ATM connections. The signalling connection control part (SCCP) [3] and transaction capability (TC) [4] provide additional information required by the call/connection to be retrieved from either a local or remote

database. The message transfer part 3 for broadband (MTP 3b) [5], the signalling ATM adaptation layer (SAAL) and the ATM layer provide for a reliable transport mechanism for signalling information between nodes within a network(s).

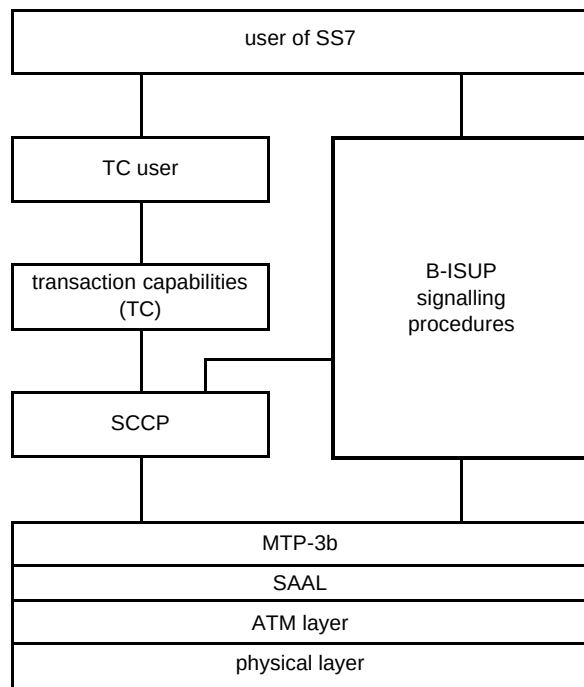


Fig 1 SS7 architecture.

3. B-ISUP signalling procedures

In the same way as N-ISUP (Q.761-4) [2], the definition of the entire B-ISUP CS-1 signalling protocol has been divided into four parts, ITU-T Recommendations Q.2761—4 [1]:

- Q.2761 — defines a functional description or overview of B-ISUP,
- Q.2762 — defines the messages and parameters applicable to B-ISUP,
- Q.2763 — defines the specific format of the messages and parameters applicable to B-ISUP,
- Q.2764 — defines the basic call/connection procedures or behaviour of B-ISUP.

3.1 B-ISUP architecture

In contrast to N-ISUP, the B-ISUP protocol architecture is based upon the concepts of the OSI application layer structure (ALS) [6]. This architecture is shown in Fig 2. The ALS technique allows the signalling procedures to be

defined in a very modular fashion, allowing easier evolution potential for future capabilities.

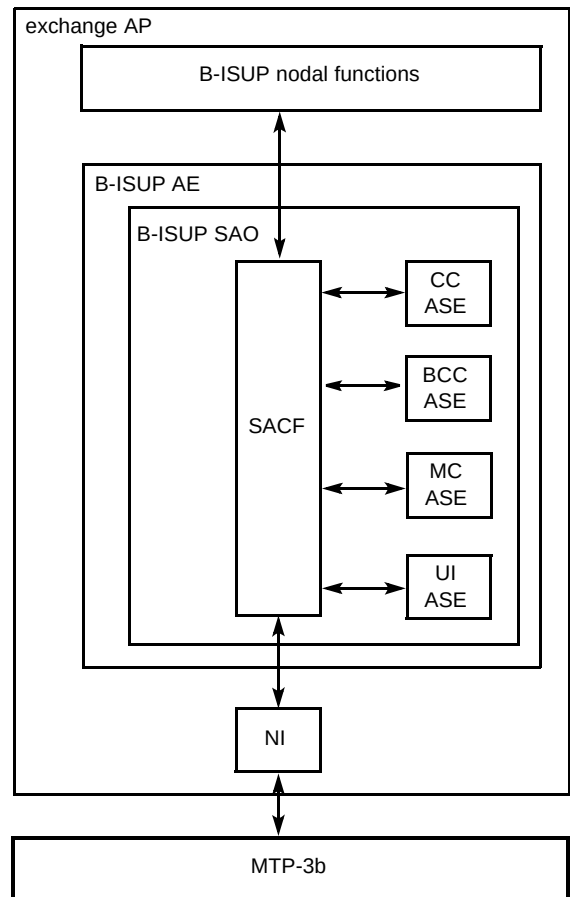


Fig 2 B-ISUP architecture.

The term 'exchange application process' is used to cover all the functionality found within a switch. The B-ISUP nodal functions and the network interface (NI) entities are static, in the sense that they are not dynamically created/deleted whenever a new call/connection is received/released.

Each application entity (AE) can contain one (or more) single association object(s) (SAOs). These SAOs contain a single association control function (SACF) and a number of individual application service elements (ASEs). The SACF acts like a co-ordinator, distributing signalling information to the correct entity, be it an ASE, the NI or an application.

ASEs

One or more ASEs can be associated with an SACF. These ASEs can be thought of as sub-routines, each designed to perform a specific function. The B-ISUP

signalling procedures consist of four basic ASEs. These are known as call control (CC), bearer connection control (BCC), maintenance control (MC) and an unrecognised information (UI).

The behaviour of an SAO is described using a formal description language known as specification and description language (SDL) [7]. SDL is based upon a finite state machine, each consisting of a number of states. A change of state occurs due to an event. An event can be a message or primitive that has been sent or received, or the expiry of a timer. In B-ISUP, once the necessary signalling information has been exchanged between the nodes in the correct sequence, according to the signalling procedures, the state machines are said to be in the 'active state'. When all the signalling nodes involved in the call/connection are in the active state, the call/connection is said to be established and user data can be transferred.

The CC and BCC ASEs contain the procedures to establish, maintain and release an ATM call and a connection simultaneously. Although the CC and BCC ASEs are logically separated, presently B-ISUP is a combined call control and bearer control protocol. This can be seen from Table 1, where some messages are related to both the CC and BCC ASE. More information on call/bearer separation is contained in Knight and Law [8]. CC and BCC ASEs consist of two parts, an incoming side and an outgoing side. Therefore a switch will have two B-ISUP AEs to support incoming and outgoing call/connections. The MC ASE handles the procedures for resource resetting, blocking, user part availability and consistency checking, while the UI ASE deals with all unrecognised message and parameter handling.

Additional B-ISUP capabilities can be incorporated by either modifying the existing ASEs or by including a new ASE(s) within an SAO.

Table 1 ASE/message relationship.

Message	Related ASEs
Initial address message	CC, BCC
Initial address message acknowledgement	BCC
Answer	CC, BCC (Note 1)
Release	CC, BCC
Release complete	BCC
Blocking	MC
Reset	MC
User part test	MC
Note 1: Depends on message parameters	

For each new call/connection, a new instance of a B-ISUP AE is created. These AEs are identified by a signalling identifier (SID). Like point codes, there are both destination and originating SIDs (DSIDs and OSIDs). When a call/connection is released, the B-ISUP AE instance is deleted and the SID value is freed. The network interface (NI) function uses the SID parameter to distribute messages received from MTP-3b to the appropriate instance of B-ISUP AE. Figure 3 shows the B-ISUP AE instances that are created for a single call/connection scenario, involving three nodes as described in section 4.4.

The association between B-ISUP messages and the related ASE(s) can be seen in Table 1.

3.2 Message structure

A B-ISUP signalling message is sent between nodes, encapsulated within the signalling information field (SIF) carried by an MTP service primitive. The format or coding

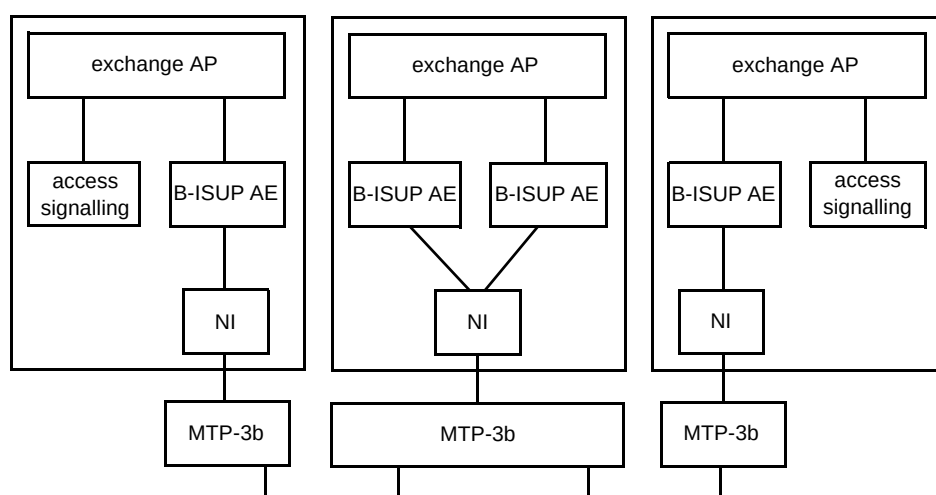
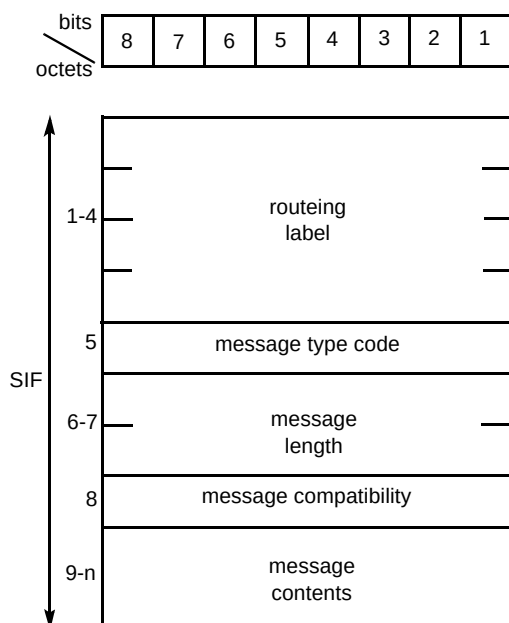


Fig 3 B-ISUP AE instances.

of the B-ISUP signalling messages is shown in Fig 4. The fields are shown coded as octet groups.



routing label: specifies originating and destination point codes, (OPC and DPC) plus the signalling link selector (SLS)

message type: identifies the type of message, IAM, IAA, ACM, etc

message length: specifies the length of the message contents; the length of this will depend upon number of parameters associated with each message

message compatibility: identifies what actions should be taken by the switch if the message is not understood (i.e. a CS-2 specific message being received by a CS-1 switch)

message content: contains the parameters related to the message type

Fig 4 Relationship between the SIF and associated parameters.

The B-ISUP CS-1 signalling procedures support 28 different signalling messages. The main signalling messages are described below:

- **IAM** — the initial address message (IAM) is the first message exchanged between the originating and destination exchanges, containing the called party number parameter, plus the required ATM characteristics, such as VPCI/VCI value, QoS class, ATM traffic descriptor, broadband bearer capability, and AAL parameters,
- **IAA** — the IAM acknowledgement message (IAA) is used as an acknowledgement of the IAM and indicates that the requested resources have been reserved for the requested call/connection,
- **IAR** — the IAM reject message (IAR) is sent in response to an IAM to indicate that the call/connection has not been accepted, due to lack of resources,

- **ACM** — the address complete message (ACM) is sent from the destination exchange towards the originating switch, to indicate that the destination switch has all the addressing information to reach the called party,
- **ANM** — the answer message (ANM) is sent from the destination switch towards the originating switch, and is used to indicate that the called party has accepted the call/connection,
- **REL** — the release (REL) message can be sent from either the originating or destination switch, to indicate that either party wishes to release the call/connection,
- **RLC** — the release complete message (RLC) is sent in response to a received REL; it is used to acknowledge that the far end has agreed to the release of the call/connection.

3.3 Parameter structure

Within the message content field, there will be a number of parameters which contain information associated to that particular message. These parameters are referred to as information elements (IEs) within broadband access signalling (ITU-T Recommendation Q.2931). The structure of the parameters is very similar to the message format, with type, length, compatibility and content fields. Some messages, have many more parameters associated with them than others. For example, the IAM message can contain over 30 parameters, while other messages contain only a single parameter. These parameters can be of either a fixed or variable length, and can be either mandatory or optional. The structure and sequence of the parameters is defined using the Abstract Syntax Notation 1 (ASN.1) as shown in ITU-T Recommendation Q.2763 [1].

When a switch receives a message, it has to locate, identify and check all the parameters contained in the message content field, to see if the necessary information is present. Based on this received information, the B-ISUP procedures determine what action should be taken, such as continue with the call/connection or reject the call/connection attempt.

3.4 QoS and ATC parameters

The ability to specify a quality of service (QoS) level associated with a connection and for the network to guarantee that quality for the lifetime of the connection is one of the major benefits of ATM. B-ISUP CS-2 will support the use of QoS classes. Four classes have been defined by the ITU-T [9]. The QoS class requested for the connection is signalled in the IAM, and once accepted cannot be changed for the lifetime of the connection. If any of the node(s) along the route cannot support the required QoS class, the connection request will be rejected. The

parameters which constitute the definition of QoS can be split into two basic categories:

- delays, which consist of cell transfer delays and variation in cell delays,
- errors, which consist of lost cells, errored cells, mis-inserted cells, and blocks of cell errors.

ATM can support a number of ATM transfer capabilities (ATCs) or service categories¹. The ATCs supported by B-ISUP include variable bit rate (VBR), constant bit rate (CBR), ATM block transfer (ABT), and available bit rate (ABR)². VBR is sub-divided into three separate conformance definitions, known as VBR.1, VBR.2 and VBR.3. VBR.1 has time constraints imposed on the cell transfer and delay variation times, whereas VBR.2 does not. VBR.3 is also known as 'tagging', whereby rather than the ATM cell being discarded by the switch if it is non-conformant, the ATM cell priority is reduced instead. A conformance definition describes the method used to determine conformant cells, against the connection traffic descriptor, and how those cells judged to be non-compliant are handled. Detailed information regarding tagging and conformance definitions can be found in ITU-T Recommendation I.371 [10].

These QoS classes, along with their respective ATCs, are shown in Table 2. Class 1 is defined as the highest QoS, while Class U is the lowest.

Table 2 Relationship between ATCs and QoS classes.

ATC	Applicable QoS class
CBR, VBR.1, ABT	Class 1 (stringent class)
CBR, VBR.1, ABT	Class 2 (tolerant class)
VBR.2, VBR.3, ABR	Class 3 (bi-level class)
Any ATC	Class U

3.5 Address parameters

B-ISUP supports the traditional E.164 numbering/addressing plan, plus additionally the ATM end-system address (AESA). The AESA consists of an E.164 number plus a network service access point (NSAP) address. Routing within a B-ISUP network is currently based on the E.164 number contained in the called party number parameter and the NSAP address is carried transparently through the network. Additional addressing schemes are likely to be included in future B-ISUP Recommendations.

¹ ITU-T refer to ATCs while ATM-F refer to service categories.

² ATM-F refer to CBR and VBR which is equivalent to ITU-T's DBR and SBR, respectively.

3.6 Signalled ATM parameters

To establish, maintain and release connections in an ATM environment, a number of ATM-related parameters need to be signalled between the switches involved in the connection. The main ATM-related parameters are described below:

- ATM traffic descriptor is used to select the required connection bandwidth in both the forward and backward directions — different values for the forward and backward directions can be specified (i.e. asymmetric),
- ATM adaptation layer (AAL) parameter is used to select which type of AAL is to be used for a connection — presently the defined AAL types are 1, 3/4 and 5, each AAL type having its own set of characteristics to be used with different ATM transfer capabilities such as CBR, VBR, ABR, etc,
- broadband bearer capability (B-BC) is used to select the bearer class and the ATM transfer capability (ATC) (ITU-T Recommendation Q.2931 [11] states which bearer classes are supported) — the ATC identifies if the user's connection shall support CBR, VBR, ABR or ABT, while the ATC type must be common across all the links making up the connection (for definitions, see the ITU-T Recommendation [10]),
- BB high/low-level information — both the BB-HLI and BB-LLI are used for compatibility checking between terminals, and as such are carried transparently across the network,
- connection identifier indicates on which VPCI/VCI the user's connection is to be established,
- QoS is used to select which QoS class is required for the user's connection,
- operations, administration and maintenance (OAM) flows are used for fault and performance management of ATM connections — the OAM F5 flow relates to the VC level, while the F4 flow relates to the VP level (ITU-T Recommendation I.610 [12] contains detailed information).

ITU-T Recommendation Q.2721 lists the parameters supported by both N-ISUP and B-ISUP.

3.7 Signalling topology

The topology of SS7 signalling networks can be split into two categories. These are known as associated and non-associated modes. Associated mode of signalling is the simplest of the two, the signalling path directly connecting to the two signalling points (SPs). The SPs are nodes A and B in the example shown in Fig 5(a). Quasi-associated signalling is a form of non-associated mode signalling. In this mode, the signalling path, and hence messages, do not flow directly between the two signalling end-points. Instead

they are routed via an intermediate point, known as a signalling transfer point (STP). STPs are packet switches which route signalling messages around the network. This is shown in Fig 5(b). Although quasi-associated is more complex than associated, it allows network operators to achieve greater flexibility and resilience. Depending on the network operators' requirements a signalling network can be configured using associated, quasi-associated or a mixture of both signalling modes.

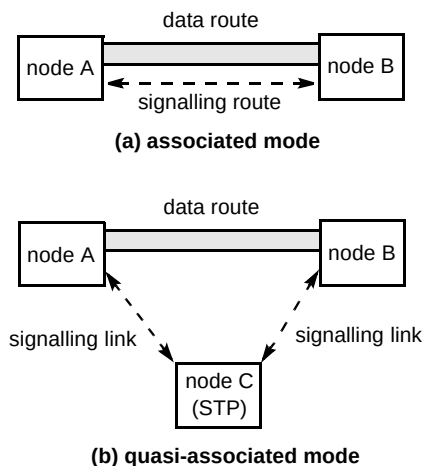


Fig 5 Signalling topology modes.

4. Routing

In an ideal network there would be no blocking, no failures of links or nodes, re-routing would never happen and the optimal route would always be chosen through the network. However, all networks suffer from such problems, and therefore the routing algorithms have to contend with these, together with the additional difficulties associated with ATM networks, where connection characteristics, and the resources allocated to them, can vary enormously.

The procedures used for routing B-ISUP signalling messages in an SS7 network is known as MTP-3b [5]. MTP-3b is based upon MTP-3 functionality³, with additional capabilities for ATM-based networks. MTP-3b uses a hop-by-hop approach to route signalling messages across a network towards the destination. Unlike source-based routing, where the source or originating switch knows the complete route through a network towards the destination, switches using a hop-by-hop approach only know how to route to the next switch nearer to the destination. The hop-by-hop mechanism does not require topology-related information to be flooded or aggregated throughout the network. Both types of routing algorithms can be used in an ATM environment and both have their own advantages and disadvantages, as explained in Scott and Jones [13].

³ With the exception of some MTP layer 3/layer 2 management processes MTP-3b functions in a very similar manner to ITU-T Recommendation Q.704 MTP layer 3.

4.1 MTP-3b introduction

Like MTP-3, MTP-3b operates in a connectionless manner. It is designed to reliably transfer B-ISUP signalling messages between nodes within an SS7 network. To achieve this, MTP-3b is able to re-route around failures, control congestion build-up of messages and detect errors, such as messages received out of sequence. The major difference between the MTP-3 and MTP-3b is concerned with the maximum length of the signalling information field (SIF) that can be transferred in a single message. With MTP-3b, this has been increased to 4091 octets to make more efficient use of the underlying SAAL capabilities, against the 272-octet limit for narrowband links. Therefore, provided the 272-octet limit is not exceeded, B-ISUP is equally able to utilise existing narrowband MTP-based networks. MTP-3b, like MTP-3, is split into two main sections, known as signalling message handling and signalling network management.

4.2 Signalling message handling

Signalling message handling is responsible for routing the user part messages to their destination, and once there, to deliver them to the intended user part. The actual routing process is based on point codes (PCs), which in the UK are 14 bits in length. Each PC uniquely identifies a node, be it in a national or international network. There are two types of PC within the routing label — destination PC (DPC) which identifies the intended destination node, and an originating PC (OPC) which identifies the originating node. ITU-T Recommendation Q.708 [14] has detailed information on PC formats. In addition, the routing label contains the signalling link selector (SLS), which is used to evenly distribute the user part messages over a number of signalling links to help avoid congestion on any one link. The structure of the routing label is shown in Fig 6.

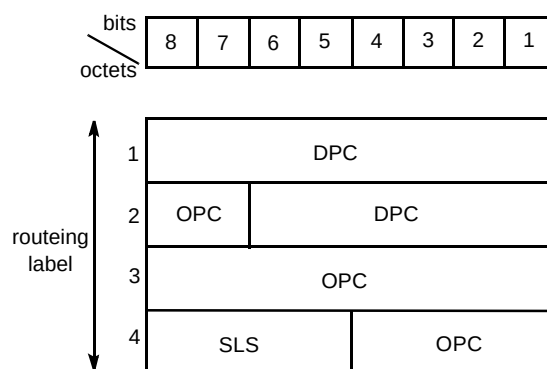


Fig 6 Routing label.

Receiving MSUs

Signalling information is transferred between MTP nodes by using message signal units (MSUs). Whenever MTP-3b receives an MSU from the SAAL, it is examined

by the discrimination function. This examines the routing label within the signalling information field (SIF) to determine the DPC value. If the DPC does not match the DPC value of the node, the node uses the route management function to consult its routing database to select the next link towards the destination. If the DPC matches that of the node, the message has reached its destination and is passed to the distribution function.

The distribution function examines the service information octet (SIO) within the MSU to determine for which user part or network management functionality the message is intended.

Sending MSUs

When MTP-3b receives a message from a higher layer, the process is the same as relaying a message on to another node. Therefore the routing process uses the DPC to access the routing tables to decide which route is to be taken and the SLS determines on which signalling link the messages are to be transmitted.

4.3 Signalling network management

The signalling network management maintains the integrity of the signalling network in the event of failure. It consists of three parts — signalling traffic management, signalling link management and signalling route management.

- Link management

This controls a signalling link set. A link set contains one or more signalling links between two points. The link management covers link and link set activation, restoration and deactivation. MTP-3b link management, to a large part, is identical to that of MTP-3 with the exception of link change-over procedures which employ different management messages, and triggers appropriate to the SAAL.

- Traffic management

This is used to divert signalling messages from one route to another without any loss, duplication or mis-sequencing and it can then divert them back to the original route when required.

- Route management

This is concerned with the status and availability of signalling routes. Information is distributed around the network in order to block or unblock particular routes from being used, such as when a node has failed and all routes to that node are blocked.

Figure 7 shows how the signalling message-handling and signalling network-management functions interact with one another.

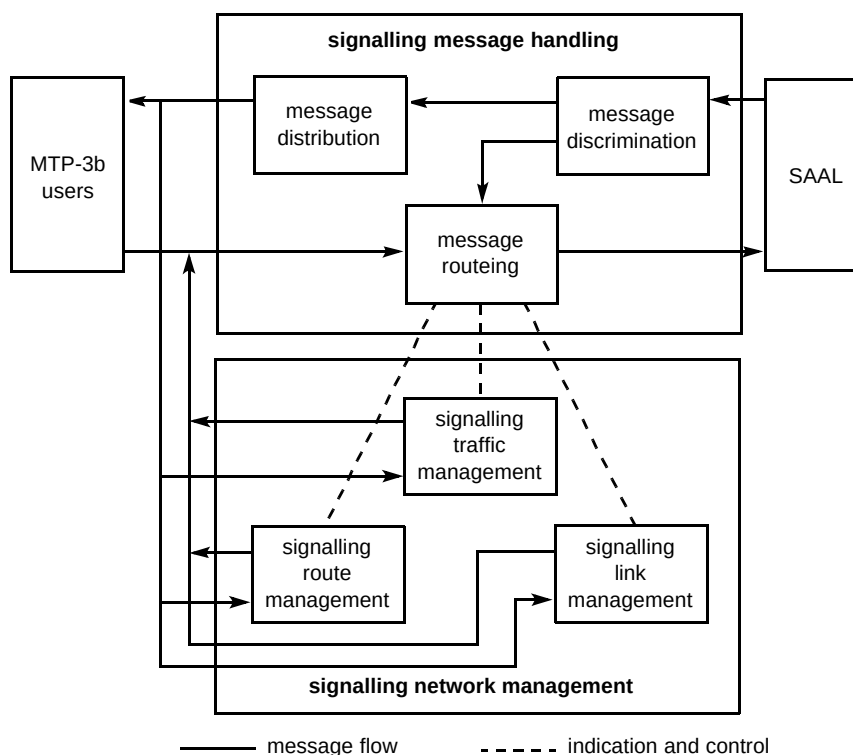


Fig 7 MTP-3b overview.

4.4 Signalling message flows

Figure 8 shows an example of the UNI and NNI signalling message flows for an *en bloc*, manually answered terminal. *En bloc* sending means that all the addressing information is sent at once, instead of in stages like overlap sending. Examples of many other possible scenarios are shown in ITU-T Recommendation Q.2650 [15].

The following gives a brief explanation of the main events in establishing and releasing an ATM call/connection. The paragraph numbering corresponds to the numbering in Fig 8.

1. B-ISUP has the concept of assigning and non-assigning procedures. For all the VPCI values which the two adjacent switches control, one will be the assigning switch for all the even-numbered values, while the other will be the assigning switch for all the odd numbered values. This procedure stops contention over VPCI/VCI values and bandwidth between adjacent switches. The assigning switch is responsible for

selecting a VPCI/VCI and reserving the required connection bandwidth on that VPCI/VCI. The information elements (IEs) contained in the UNI SETUP message are processed by the originating switch. Assuming this switch is the assigning switch, and the connection admission control (CAC) has accepted the call/connection, the DPC is calculated from the called party number and a VPCI/VCI value chosen. Selection of the route can depend upon the called party number, plus ATM cell rate, broadband bearer capability, QoS and associated delay parameters. The switch will create a new instance of the B-ISUP AE, as described in section 2. The relevant IEs within the SETUP message, plus the necessary NNI parameters, are transferred within the IAM towards the terminating switch. The actions performed by the two other switches will depend on them acting as the assigning or non-assigning switches. Both of these switches will create an instance of the B-ISUP AE, providing they can accept the call/connection. The IAA is sent back as an acknowledgement that the IAM has been accepted.

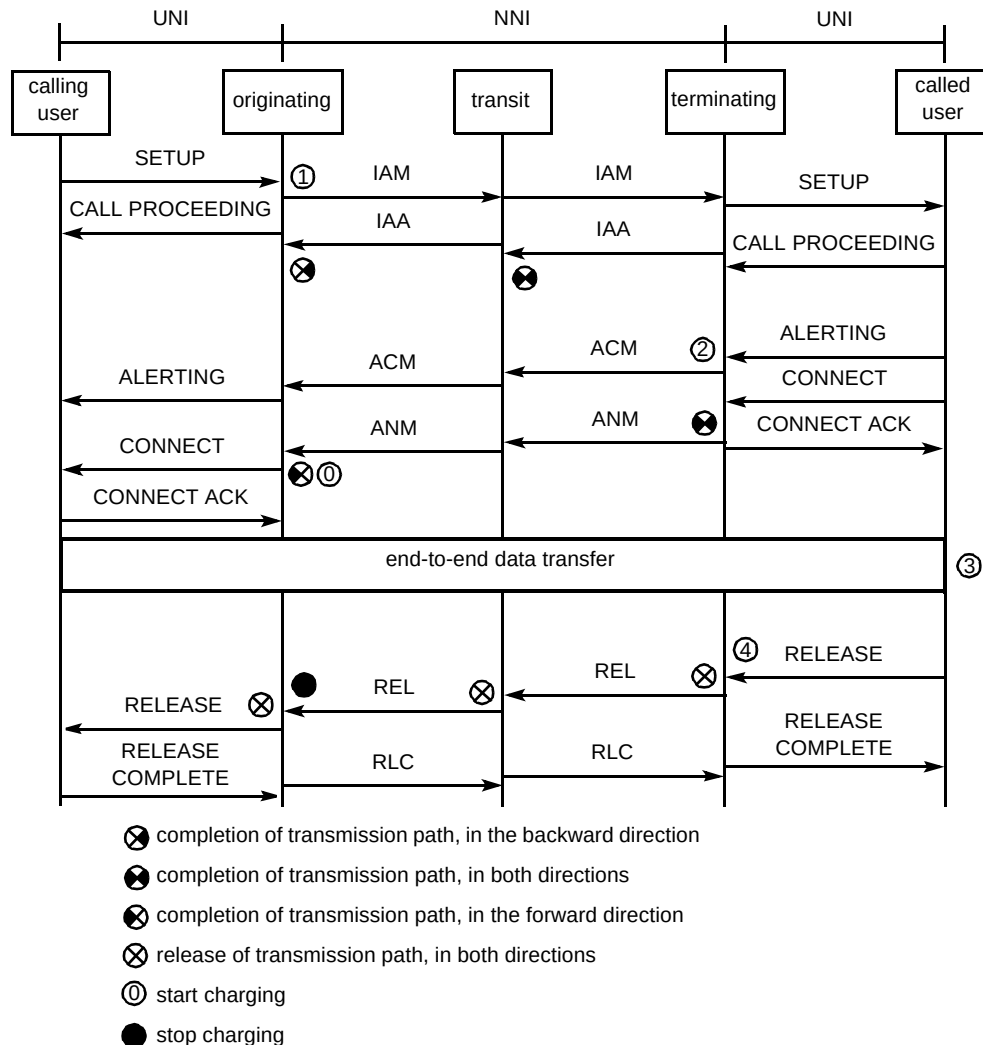


Fig 8 End-to-end message flows.

2. When the terminating switch determines that the called party is being alerted, it concludes that all the necessary address information is complete and sends an ACM back towards the originating switch. If the called party responds with a CONNECT, identifying that the call/connection has been accepted, an ANM is sent back towards the originating switch. These B-ISUP messages are routed back to the appropriate B-ISUP AE instance within each switch along the route by the combination of the DPC, SIO and SID identifiers.
3. Once the connection has been established, charging is initiated and user plane information can be exchanged, between the two end users. With connection-oriented networks, a substantial amount of processing has to be carried out to establish the connection. If, subsequently, the connection is only maintained for a short period of time, or if only a small amount of user data needs to be transferred, this represents an inefficient use of resources. Policing of the user plane ATM cells is employed by the switches to ensure the traffic contact agreed at connection set-up time, via the signalling messages, is not violated. Any user plane cells judged to be in breach of this contract are either discarded or tagged (i.e. given a lower priority).
4. B-ISDN access signalling, Q.2931 [11], supports a two-phase release procedure instead of the three-phase employed by ISDN access signalling, Q.931 [16]. Either the calling or called party can release the connection. In this example the called party initiates the release. Once the RLC message has been received and processed, all resources associated with the call/connection are freed, including the VPCI/VCI value. In addition, charging is terminated and the BISUP AE instances are deleted.

5. SCCP and TC

MTP-3b was developed to allow signalling messages of greater length to be transferred via the SAAL links with potentially much higher data rates than the MTP level 2. To make use of the new MTP-3b capabilities, SCCP has been modified so that it can allow for the transportation of up to 3952 octets of data without invoking any segmentation procedures. Segmentation is used when data is too large to be carried by an individual message, therefore the data is to be split or segmented into a number of parts and carried in multiple messages.

B-ISUP uses the functionality provided by TC and SCCP mainly for non-call-related applications. Typical applications include services such as look-ahead (Q.2724.1) which allows a network to check the called party terminal availability and compatibility without any commitment of network resources, i.e. establishing a user plane connection.

In addition, closed user group (CUG) (Q.2735) and multi-connection (Q.2722.2) can also make use of the TC and SCCP functionality. With the integration of B-ISUP and intelligent network application part (INAP) protocols planned for ITU-T CS-3 plus the development of broadband mobile networks and terminals, the functionality offered by TC and SCCP will become more important — for more information, see the appropriate ITU-T Recommendations [3, 4].

6. SAAL

In an ATM environment, MTP layers 1 and 2 are replaced by the ATM layer and signalling ATM adaptation layer, respectively. This is shown in Fig 9. As its name implies, the SAAL is the adaptation layer for the control plane (i.e. signalling information), whereas the various AAL types are the adaptation layers for the user plane (i.e. user data). The SAAL consists of two parts, the common part and the service-specific part. The common part is based upon AAL5, which is also one of the user plane AAL types. The purpose of AAL5 within the SAAL is to segment the signalling messages into ATM cells of 53 bytes, ready for transportation by the ATM layer and also to re-assemble the ATM cells back into complete signalling messages. The service-specific part consists of the service-specific co-ordination function (SSCF) and the service-specific connection-oriented protocol (SCCOP). These are used to transfer variable-length packets, known as service data units (SDUs), which contain signalling information. These SDUs can be transmitted between nodes using either an assured (i.e. using error detection, flow control) or unassured manner.

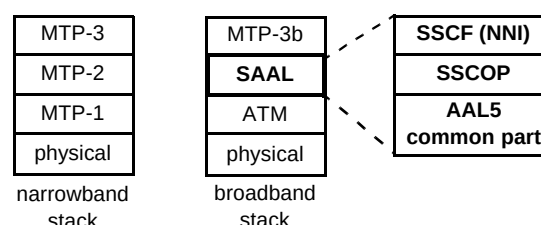


Fig 9 Narrowband and broadband protocol stacks

7. Interworking

7.1 Access signalling

The four main ATM UNI or access signalling protocols are the ITU-T's Q.2931 [11] B-QSIG [17] plus the ATM-Forum's UNI v3.1 [18] and UNI signalling v4.0 [19]. As B-ISUP and Q.2931 are defined by the same organisation, the ITU-T, interworking between the two is considered paramount and both groups work together to achieve this. ITU-T Recommendation Q.2650 defines the necessary procedures for NNI and UNI integration. Interworking protocols from different organisations is always a problem as different organisations have different requirements. Interworking with UNI v3.1 causes problems as UNI v3.1 only supports automatically answered terminals (i.e. it has no

ALERTING message). UNI v4.0 supports the ALERTING message, plus other additional signalling capabilities. Some of the capabilities include individual QoS parameters instead of just QoS classes, anycast, available bit rate (ABR), leaf-initiated join (LIJ), generic ID transport (GIT), proxy signalling concept, point-to-multipoint connections and VP switching. B-ISUP CS-2 either presently supports or is planning to support these additional UNI v4.0 capabilities, except proxy signalling (which is UNI specific), QoS parameters and anycast.

7.2 PNNI

The interworking between B-ISUP and the ATM-F's private network/network interface (PNNI) [20] is seen as very important and work is being carried out within the ATM-F to address this issue. The ATM-F is developing the ATM internetwork interface (AINI), which is explained in more detail in Scott and Jones [13].

7.3 B-ICI

Broadband inter-carrier interface (B-ICI) was defined by the ATM-F, to be used for interconnecting operators' networks. B-ICI v1.0 was based upon permanent VCs (PVCs) and thus had no signalling capabilities. B-ICI v2.0, [21] is based upon B-ISUP, using a subset of MTP-3b allowing for only associated mode signalling (i.e. no support STPs). Neither the ITU-T nor ATM-F has published an interworking specification — as far as B-ISUP and B-ICI are concerned, a reasonable level of interworking should be possible as the two protocols are similar. Recently, the interest in the future development of B-ICI are concerned, has slowed down and plans for B-ICI v3.0 have been frozen, as effort is being concentrated on the development of AINI.

7.4 N-ISUP

B-ISUP supports the necessary procedures, messages and parameters to achieve interworking with N-ISUP. Various B-ISUP and N-ISUP interworking scenarios are defined in ITU-T Recommendation Q.2660. Interworking at layer 2, between MTP-3 and MTP-3b can be achieved, as long as the information carried by MTP-3b is limited to 256 octets in length.

8. B-ISUP capabilities

The capabilities of the different B-ISUP sets are summarised below and in Table 3.

8.1 CS-1

CS-1 supports basic point-to-point call/connections, along with the traditional supplementary services.

8.2 CS-2

CS-2 supports the additional ATCs, plus the modification and negotiation of connection characteristics. More advanced connection topologies, such as point-to-

multipoint and multi-connection are also being developed along with switched virtual paths, soft permanent virtual connections and leaf-initiated join.

8.3 CS-3

The signalling requirements for B-ISUP CS-3 are still being defined. However, it is likely to support integration with both IN and mobile networks, along with advanced connection topologies, such as multipoint, multi-connection. It should be noted, that these advanced features within CS-3 will not be available for some time yet. For more information regarding future signalling requirements, see Bruce and Clark [22].

Some of the Recommendations are still being developed. The status of the Recommendations shown in Table 3 are relevant for mid-1997; the extra capabilities are only delta documents on Q.2761—4.

Table 3 Summary of B-ISUP capabilities.

(CS)	Recommendation title	Rec No	Status
CS-1			
	Basic call/connection	Q.2761-4	Approved
	Supplementary services (MSN, DDI, CLIP, CLIR, UUS, etc)	Q.2730	Approved
CS-2			
	Generic concepts	Q.2721	Approved
	Additional traffic parameters	Q.2723.1	Approved
	ATC	Q.2723.2	Approved
	ABR	Q.2723.3	Approved
	ABT	Q.2723.4	Approved
	CDVT	Q.2723.5	Frozen
	Tagging	Q.2723.6	Frozen
	AESA	Q.2761.1	Approved
	Negotiation	Q.2725.1	Approved
	PCR modification	Q.2725.2	Approved
	PCR+SCR+MBS modification	Q.2725.3	Approved
	Modification with negotiation	Q.2725.4	Draft
	Look-ahead	Q.2724.1	Approved
	CUG	Q.2735	Approved
	Frame relay	Q.2727	Approved
	Point-to-multipoint	Q.2722.1	Approved
	Multiconnection	Q.2722.2	Draft
	User generic ID/GIT	Q.2726.4	Approved
	Priority call	Q.2726.2	Approved
	Network call correlation ID	Q.2726.3	Approved
	Switched virtual paths	NA	Frozen
	Soft PVP/Cs	NA	Frozen
	Leaf-initiated join	NA	Draft
	3rd party set-up	NA	Not started
CS-3			
	Support of intelligent networks (IN)	NA	Not started
	Support of mobility	NA	Not started
	Multipoint, multiconnections	NA	Not started
	Support of session control	NA	Not started

9. Conclusions

Since 1994 when B-ISUP CS-1 was defined, it has made significant progress, by supporting many more capabilities. In the near future, B-ISUP is set to extend its capabilities further, as it attempts to develop new addressing schemes, and work with the Internet Engineering Task Force (IETF) to further IP and ATM integration, and with the ATM-F to interwork with their respective protocols. Also, new call control proposals have been defined, which separate the call and connection aspects of the present B-ISUP protocol, making the development of CS-3 requirements easier.

The next few years will see more network operators deploying their initial ATM networks with signalling capabilities. It will be interesting to see which internodal signalling protocol, be it B-ISUP or PNNI, they choose to use, as both have their strengths and weaknesses and operators will be forced to make a difficult strategic decision.

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Ian Jones joined BT in 1994 after graduating from Aston University, with an MSc in Telecommunications Technology. After joining, he worked on producing software models to demonstrate the behaviour of broadband internodal and access signalling protocols.

He now works on defining the future signalling capabilities related to BT's CellStream and CBP programmes. He represents BT at the ITU-T B-ISUP standardisation meetings and also at the UK PNOIG ATM task group committee.